Lecture 11 (Simplex Method)

February 24, 2026

Today we will start talking about the Simplex Method. It will take a couple of class periods to fully describe it and prove some related results.

Description

In a nutshell, here is how simplex method works. Start with some BFS, $\vec{v}$. Then check each edge containing $\vec{v}$ as one of its endpoints. If you move along this edge and the value of the objective function improves, we call this edge an **improving edge**. We have 3 cases.

1. No improving edges exist (This means $\vec{v}$ is optimal and we terminate. Recall for convex functions locally optimal points are globally optimal).
2. There exists an improving edge that is also a cone (This means the LP is unbounded and no optimal solution exists).
3. There exists an improving edge with endpoints, $\vec{v}$ and $\vec{v}^+$. In this case, replace $\vec{v}$ with $\vec{v}^+$ and repeat this process.

Most of the challenge with simplex method comes from knowing how to cleverly select the right improving edge and determining when an edge is an improving edge. We will build a few preliminaries today, and use these to describe the full algorithm in the future.

Preliminaries

Recall every LP can be written in standard form:

$$\min_{\vec{x}\in\mathbb{R}^n} \vec{c}\cdot\vec{x}$$

s.t.

$$A\vec{x} = \vec{b}$$

$$\vec{x} \geq 0$$

for some $A \in \mathbb{R}^{m\times n}$.The feasible region is a polyhedron that can be written in standard form as well:

$$P = \left\{\vec{x} | A\vec{x} = \vec{b}, \vec{x} \geq 0\right\}$$

When deriving some results, we will assume that $A\vec{x} = \vec{b}$ has a solution. This can be checked by row reducing. We also are assuming that $m \leq n$ and that $\text{Rank}\,(A) = m$. If this isn't the case, we simply remove any rows that do not contain any leading ones when row reducing. Use this new system for A. These correspond to redundant constraints.

Feasible Directions

Definition

Let P be in standard form. Then $\vec{d} \in \mathbb{R}^n$ is a **feasible direction at** $\vec{x}$ if $\exists\theta > 0$ such that $\vec{x} + \theta\vec{d} \in P$. This means we can take a step of size θ in direction $\vec{d}$ from $\vec{x}$ and we will remain in P.

Some Notation

Suppose $\vec{x} \in P$ is a BFS and let $B(1), \ldots, B(m)$ be the indices of the basic variables. For convenience, let A_i correspond to the ith column of A. Let

$$B = \begin{bmatrix} A_{B(1)} & A_{B(2)} & \ldots & A_{B(m)} \end{bmatrix}$$

be the matrix of active columns. We also will have $\vec{x}_B = \begin{pmatrix} x_{B(1)} & x_{B(2)} & \ldots & x_{B(m)} \end{pmatrix}$ as the vector of active variables.

Results

As $\vec{x}$ is a BFS, we know that B must have n LI constraints for rows. This implies that B is invertible. Further, as $\vec{x}$ is feasible, we see

$$A\vec{x} = \vec{b},\ \vec{x} \geq 0$$

and as every non-basic variable must be 0 in $\vec{x}$, we see

$$A\vec{x} = B\vec{x}_B = \vec{b},\ \vec{x}_B \geq 0$$

So

$$\vec{x}_B = B^{-1}\vec{b}$$

Suppose we select a non-feasible variable, x_j. Then we can move towards x_j by using the vector $\vec{d}$ with the requirement

$$d_i = \begin{cases} 1 & i = j \\ 0 & \text{if } i \neq j \text{ and } x_i \text{ is not basic} \end{cases}$$

If we want to remain in the feasible region, we need to set additional requirements on $\vec{d}$. We can consider

$$\vec{d}_B = \begin{pmatrix} d_{B(1)} & d_{B(2)} & \ldots & d_{B(m)} \end{pmatrix}$$

Then considering $\vec{x} + \theta\vec{d}$ with $\theta > 0$, we see

$$A\left(\vec{x} + \theta\vec{d}\right) = A\vec{x} + \theta A\vec{d} = \vec{b}$$

But as

$$A\vec{x} = \vec{b}$$

then

$$\vec{b} + \theta A\vec{d} = \vec{b}$$

$$\theta A\vec{d} = 0$$

and as $\theta > 0$, we see

$$A\vec{d} = 0$$

And as every entry of $\vec{d}$ corresponding to a non-basic variable besides j must be 0, we can see

$$0 = A\vec{d} = \sum_{i=1}^{n} A_i d_i = \sum_{i=1}^{m} A_{B(i)} d_{B(i)} + A_j d_j$$

But as $d_j = 1$, we see

$$= \sum_{i=1}^{m} A_{B(i)} d_{B(i)} + A_j = B\vec{d}_B + A_j$$

And thus

$$B\vec{d}_B = -A_j$$

$$\vec{d}_B = -B^{-1}A_j$$

We will call $\vec{d}$ the jth basic direction. If we move toward $\vec{d}$, we remain feasible. Now in order to ensure positivity constraints hold, we need to consider two cases:

a) If $\vec{x}$ is non-degenerate, then $\vec{x}_B > 0$. This means for small enough θ, we see $\vec{x}_B + \theta\vec{d}_B \geq 0$. This implies that positivity of the variables is maintained and $\vec{d}$ is in fact a feasible direction.

b) If $\vec{x}$ is degenerate, then there exists $B(i)$ such that $x_{B(i)} = 0$. In this case, we need to be careful as $x_{B(i)} + \theta d_{B(i)}$ could be negative. We will not worry about this case yet, but we will eventually.

Reduced Cost

How does our cost change as we move in direction $\vec{d}$? We can find this! The objective function is

$$\vec{c} \cdot \vec{x}$$

thus

$$\nabla (\vec{c} \cdot \vec{x}) = \vec{c}$$

We can get our rate of change moving in direction $\vec{d}$ by taking the dot product of $\vec{d}$ with $\vec{c}$. As with above, we let

$$\vec{c}_B = \begin{pmatrix} c_{B(1)} & c_{B(2)} & \dots & c_{B(m)} \end{pmatrix}$$

So if we see

$$\vec{c} \cdot \vec{d} = \vec{c}_B \cdot \vec{d}_B + c_j d_j = \vec{c}_B \cdot \vec{d}_B + c_j$$

From our previous derivation, we saw $\vec{d}_B = -B^{-1}A_j$, thus

$$\vec{c}_B \cdot \vec{d}_B + c_j = c_j - B^{-1}A_j = \bar{c}_j$$

We call $\bar{c}_j$ the **reduced cost of** $\vec{x}_j$.

Example

We have defined a lot of terms. Let's give a specific example.

$$\text{minimize } c_1x_1 + c_2x_2 + c_3x_3 + c_4x_4$$

s.t.

$$x_1 + x_2 + x_3 + x_4 = 2$$

$$2x_1 + 3x_3 + 4x_4 = 2$$

$$x_1, x_2, x_3, x_4 \geq 0$$

This yields

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 0 & 3 & 4 \end{pmatrix}, \ \vec{b} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

Suppose we select x_1 and x_2 to be active. This means $x_3, x_4 = 0$. Further,

$$B = \begin{pmatrix} A_1 & A_2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix}$$

To solve for x_1, x_2, we take

$$B\vec{x}_B = \vec{b}$$

$$\vec{x}_B = B^{-1}\vec{b} = \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix}^{-1} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 & \frac{1}{2} \\ 1 & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Thus

$$\vec{x} = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

Suppose we wish to move towards x_3. Then $d_3 = 1, d_4 = 0$. To get the rest of $\vec{d}$ we take

$$\vec{d_B} = -B^{-1}A_3 = -\begin{pmatrix} 0 & \frac{1}{2} \\ 1 & -\frac{1}{2} \end{pmatrix}\begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} -\frac{3}{2} \\ \frac{1}{2} \end{pmatrix}$$

We see

$$\vec{d} = \begin{pmatrix} -\frac{3}{2} \\ \frac{1}{2} \\ 1 \\ 0 \end{pmatrix}$$

And we can compute the reduced cost as

$$\vec{c} \cdot \vec{d} = -\frac{3}{2}c_1 + \frac{1}{2}c_2 + c_3 = c_3 - \left(B^{-1}A_3\right) \cdot \vec{c}_B = \overline{c}_3$$

Final Results

Suppose we compute the reduced cost for a basic variable? We see $B^{-1}B = I$, so $B^{-1}A_{B(i)} = e_{B(i)}$ for all basic variable indices, $B(i)$. This implies

$$\overline{c}_{B(i)} = c_{B(i)} - \left(B^{-1}A_{B(i)}\right) \cdot \vec{c}_B = c_{B(i)} - e_{B(i)} \cdot \vec{c}_B = c_{B(i)} - c_{B(i)} = 0$$

So the reduced cost of basic variables is 0!

Theorem

Consider the BFS $\vec{x}$ associated with basis matrix, B. Let $\vec{c}$ be the corresponding vector of reduced costs. Then
a) If $\overline{c} \geq 0$, then $\vec{x}$ is optimal.
b) If $\vec{x}$ is optimal and non-degenerate, then $\overline{c} \geq 0$

Proof

a) If $\overline{c} \geq 0$, then suppose $\vec{y}$ is an arbitrary feasible solution. Let $\vec{d} = \vec{y} - \vec{x}$. Then as $\vec{x}, \vec{y}$ are both feasible, we see

$$A\vec{x} = A\vec{y} = \vec{b}$$

thus

$$A\vec{d} = A\left(\vec{y} - \vec{x}\right) = A\vec{y} - A\vec{x} = 0 - 0 = 0$$

This implies

$$A\vec{d} = B\vec{d_B} + \sum_{i \in N} A_i\vec{d_i} = 0$$

(Where N is the set of all indices corresponding to non-basic variables). Then as B is invertible, we see

$$B\vec{d_B} + \sum_{i \in N} A_i d_i = 0$$

$$B\vec{d_B} = -\sum_{i \in N} A_i d_i$$

$$\vec{d_B} = -B^{-1}\sum_{i \in N} A_i d_i = -\sum_{i \in N} B^{-1}A_i d_i$$

And we also have

$$\vec{c} \cdot \vec{d} = \vec{c}_B \cdot \vec{d}_B + \sum_{i \in N} c_i d_i = \vec{c}_B \cdot \left(- \sum_{i \in N} B^{-1} A_i \vec{d} \right) + \sum_{i \in N} c_i d_i$$

$$= - \left(\sum_{i \in N} \vec{c}_B \cdot \left(B^{-1} A_i d_i \right) \right) + \sum_{i \in N} c_i d_i = \sum_{i \in N} c_i d_i - \vec{c}_B \cdot \left(B^{-1} A_i d_i \right) = \sum_{i \in N} \left(c_i - \vec{c}_B \cdot \left(B^{-1} A_i \right) \right) d_i$$

But each $c_i - \vec{c}_B \cdot \left(B^{-1} A_i \right) = \bar{c}_i \geq 0$. As $\vec{y}$ is feasible and each $i \in N$ is non-basic, then $d_i \geq 0$ for all $i \in N$, thus

$$\left(\sum_{i \in N} c_i - \vec{c}_B \cdot \left(B^{-1} A_i \right) \right) d_i = \sum_{i \in N} \bar{c}_i d_i \geq 0$$

So moving towards $\vec{y}$ will not decrease the objective. This implies $\vec{x}$ is optimal (but not necessarily a unique optimal solution).

b) We will use contrapositive. Suppose $\vec{x}$ is a non-degenerate BFS and $\bar{c}_j < 0$ for some j. Then we know that x_j must be a non-basic variable. This is because any basic variable has a reduced cost of 0. If we move towards x_j with a non-degenerate $\vec{x}$, then we will decrease the objective function. This means that $\vec{x}$ is not optimal. (As a side note, we may have cases where $\vec{x}$ is degenerate where this isn't true. We will deal with these special cases later).